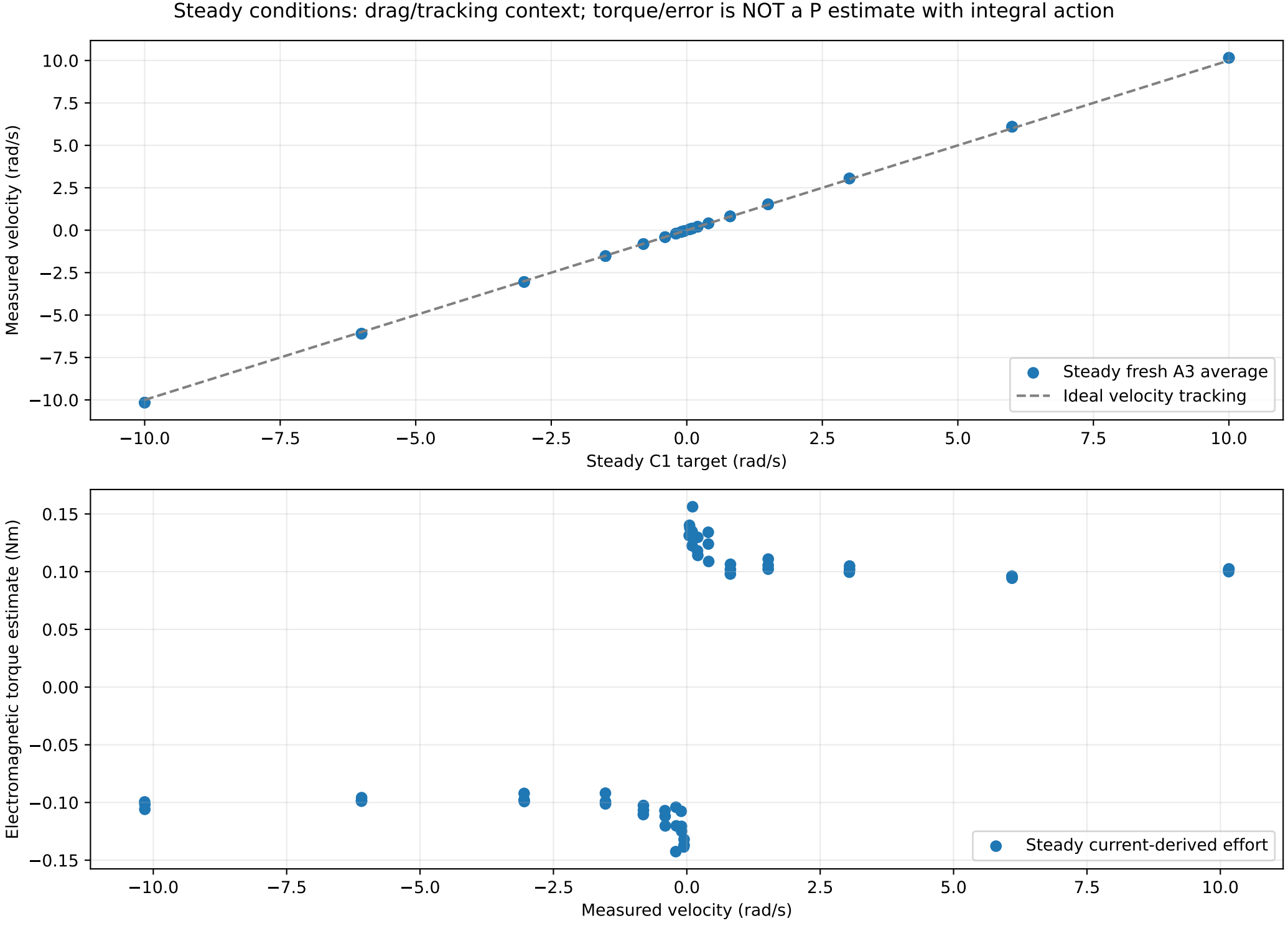
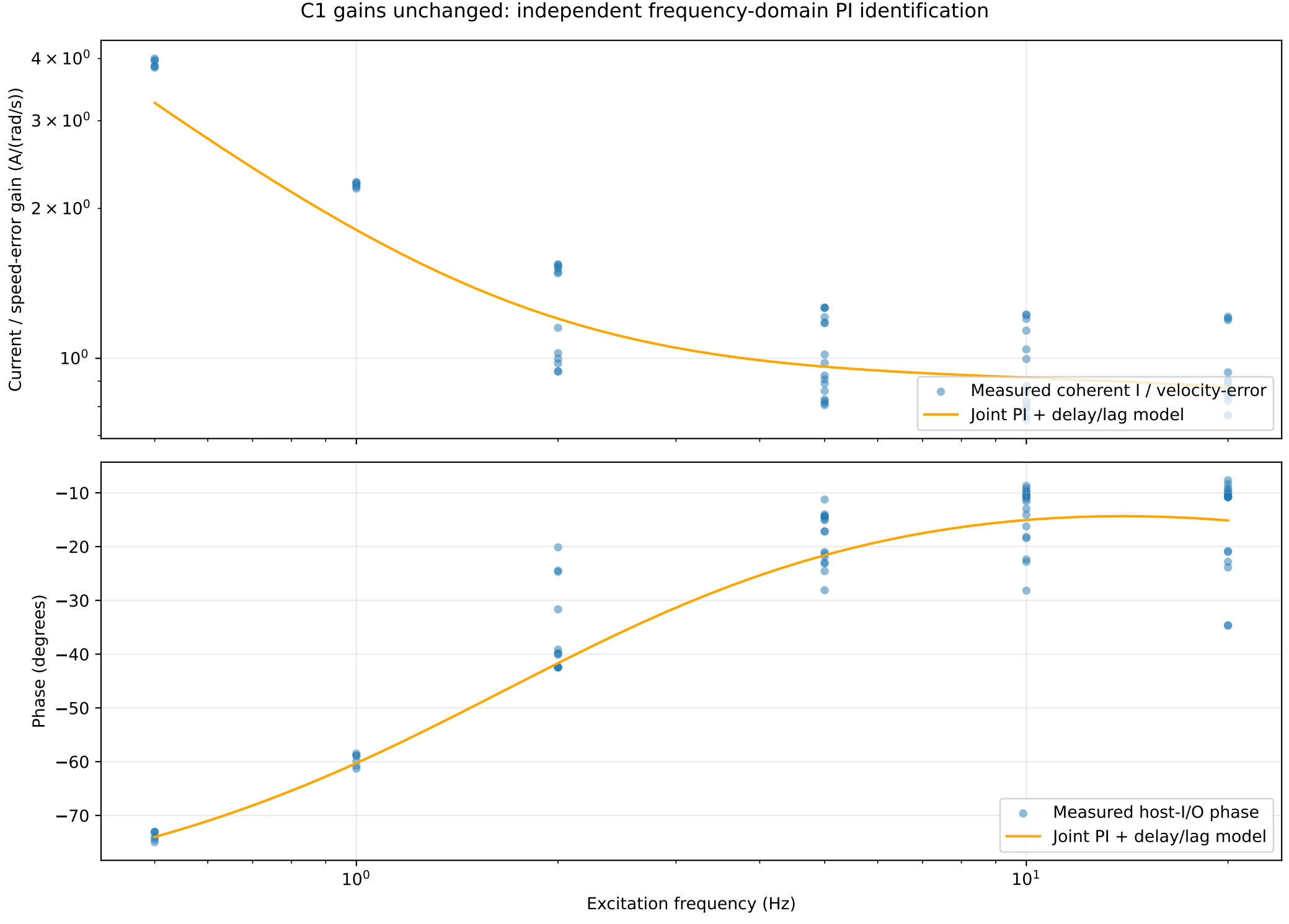
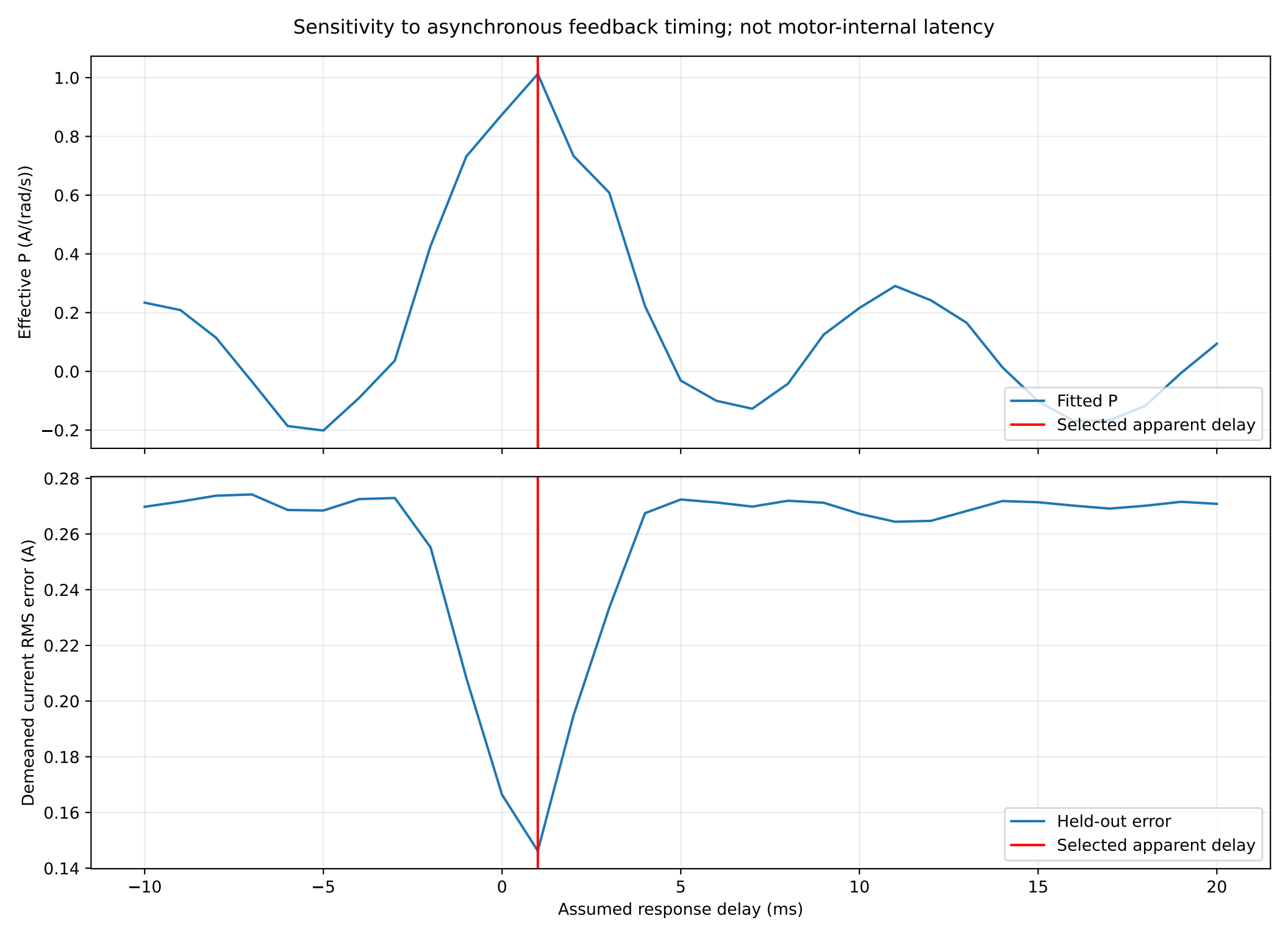
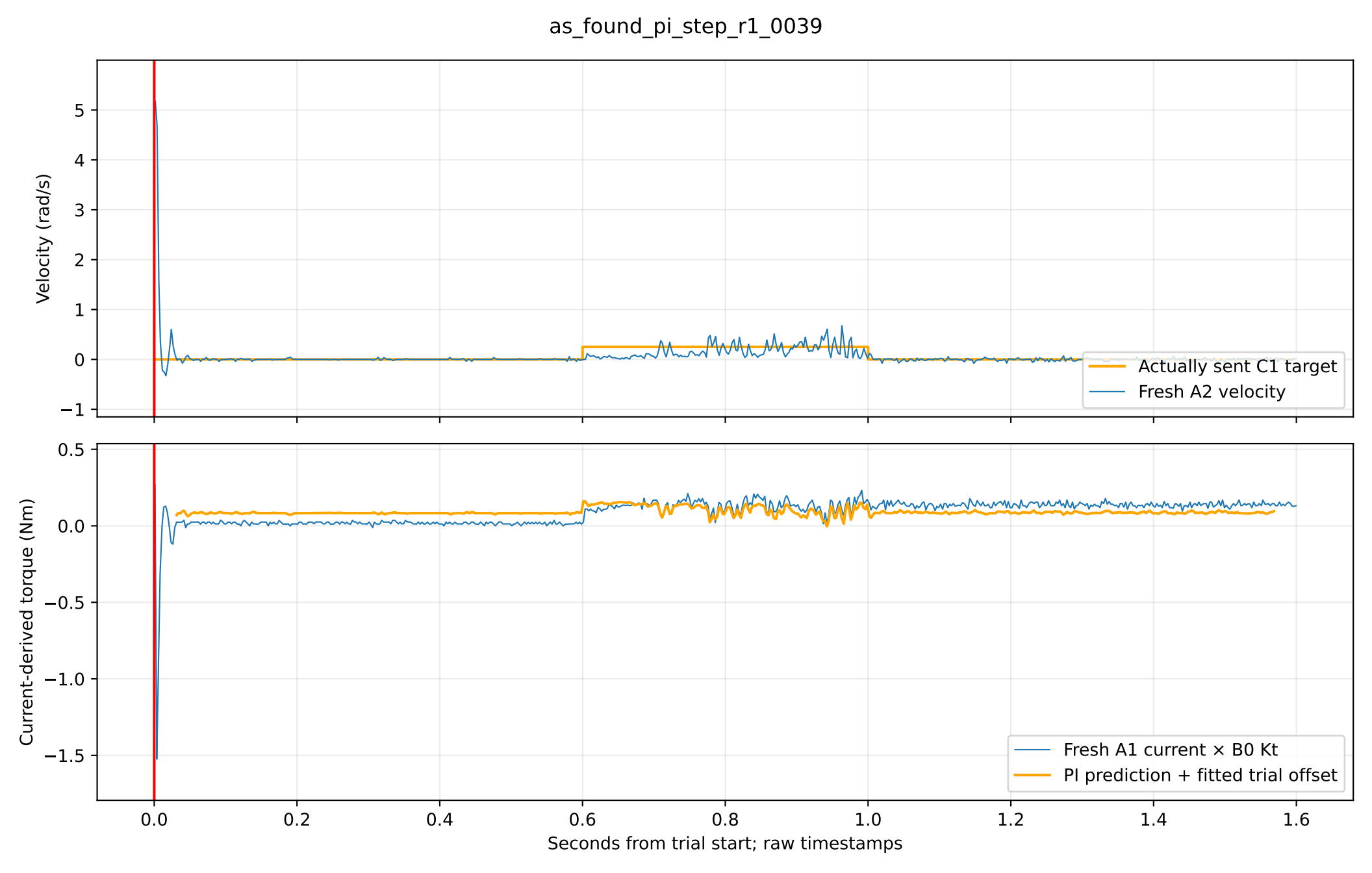
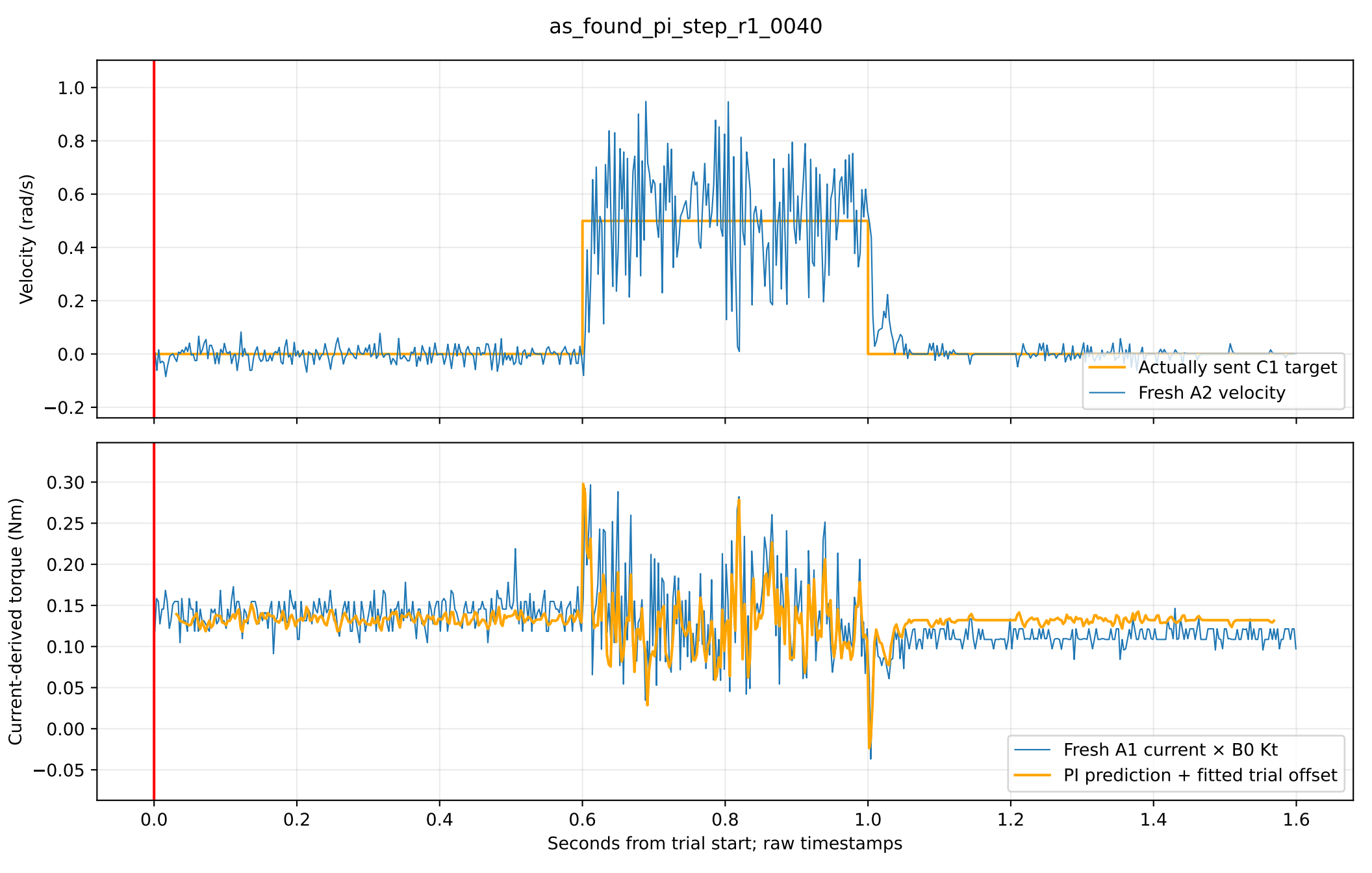
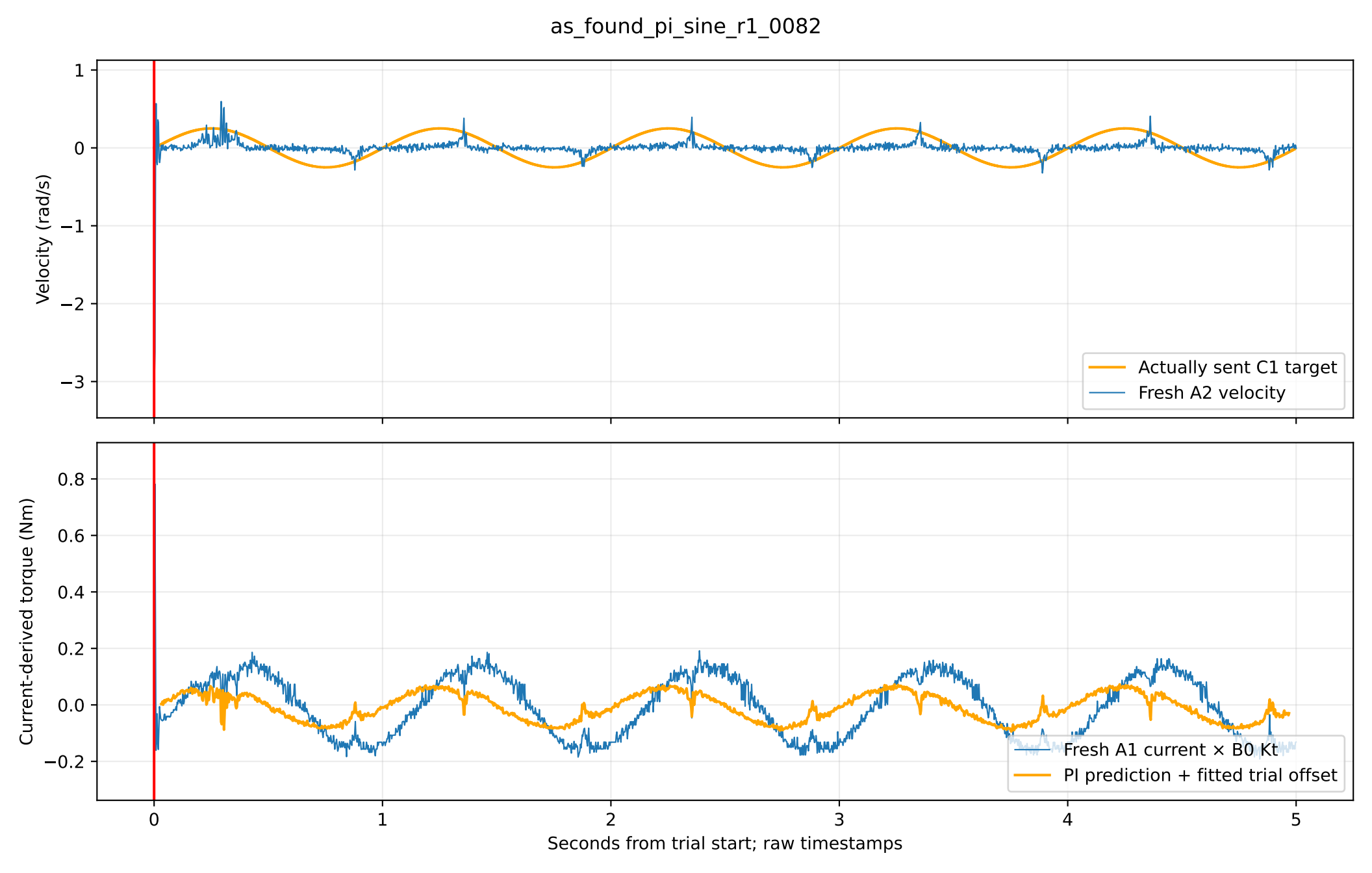

Steady conditions: drag/tracking context; torque/error is NOT a P estimate with integral action
Measured velocity (rad/s)
Steady C1 target (rad/s)
Steady fresh A3 average
Ideal velocity tracking
Electromagnetic torque estimate (Nm)
Measured velocity (rad/s)
Steady current-derived effort

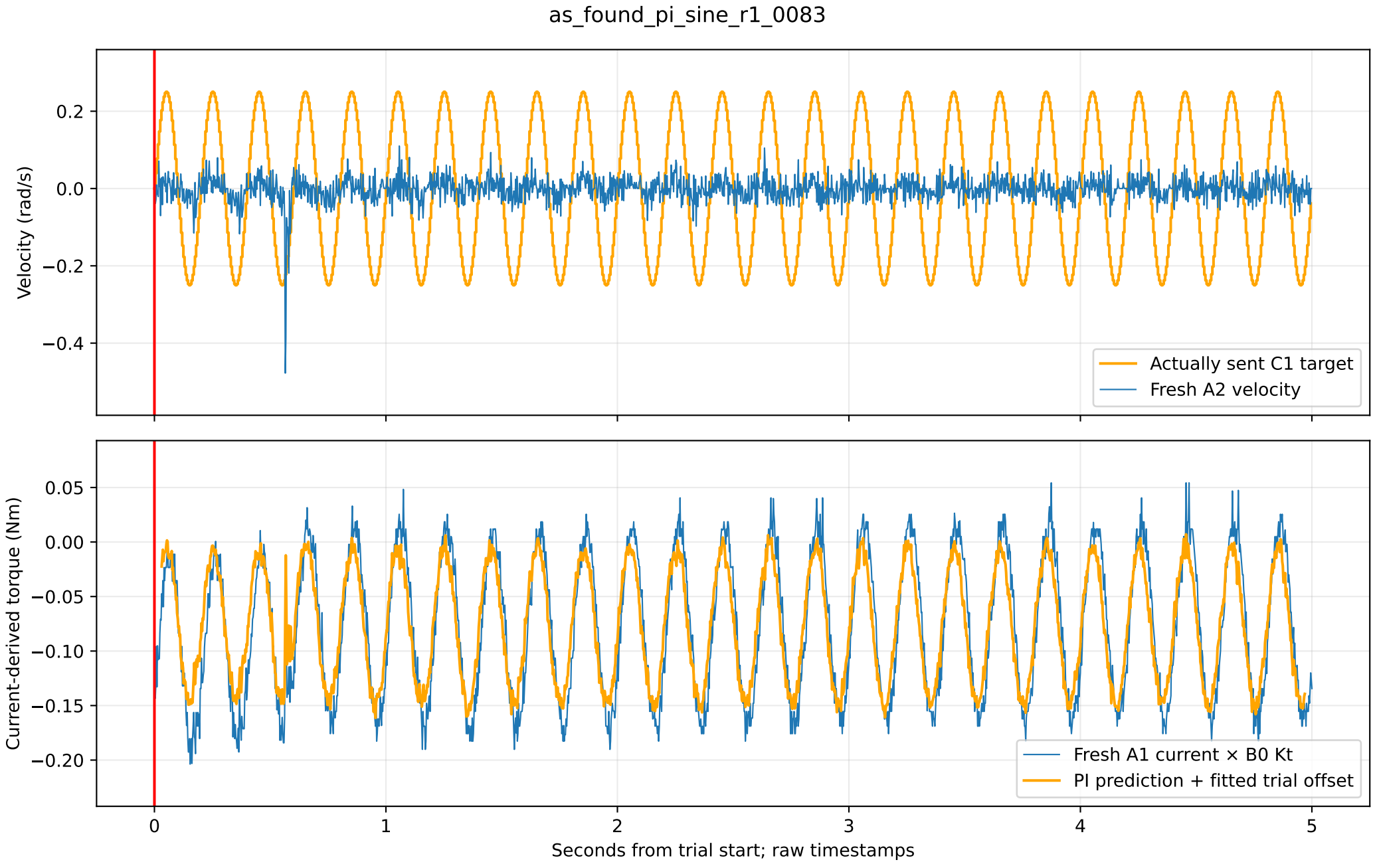

C1 gains unchanged: independent frequency-domain PI identification
Current / speed-error gain (A/(rad/s))
4×10^0
3×10^0
2×10^0
10^0
Measured coherent I / velocity-error
Joint PI + delay/lag model
Phase (degrees)
−10
−20
−30
−40
−50
−60
−70
Measured host-I/O phase
Joint PI + delay/lag model
10^0
10^1
Excitation frequency (Hz)


Sensitivity to asynchronous feedback timing; not motor-internal latency
Effective P (A/(rad/s))
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Fitted P
Selected apparent delay
Demeaned current RMS error (A)
0.28
0.26
0.24
0.22
0.20
0.18
0.16
0.14
Held-out error
Selected apparent delay
−10
−5
0
5
10
15
20
Assumed response delay (ms)


as_found_pi_step_r1_0039
Velocity (rad/s)
Actually sent C1 target
Fresh A2 velocity
Current-derived torque (Nm)
Fresh A1 current × B0 Kt
PI prediction + fitted trial offset
Seconds from trial start; raw timestamps


as_found_pi_step_r1_0040
Velocity (rad/s)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Actually sent C1 target
Fresh A2 velocity
Current-derived torque (Nm)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
Fresh A1 current × B0 Kt
PI prediction + fitted trial offset
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Seconds from trial start; raw timestamps


as_found_pi_sine_r1_0082
Velocity (rad/s)
Actually sent C1 target
Fresh A2 velocity
Current-derived torque (Nm)
Fresh A1 current × B0 Kt
PI prediction + fitted trial offset
Seconds from trial start; raw timestamps


as_found_pi_sine_r1_0083
Velocity (rad/s)
0.2
0.0
−0.2
−0.4
Actually sent C1 target
Fresh A2 velocity
Current-derived torque (Nm)
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Fresh A1 current × B0 Kt
PI prediction + fitted trial offset
0
1
2
3
4
5
Seconds from trial start; raw timestamps